

ARTEFACT 1 REPORT (A1)

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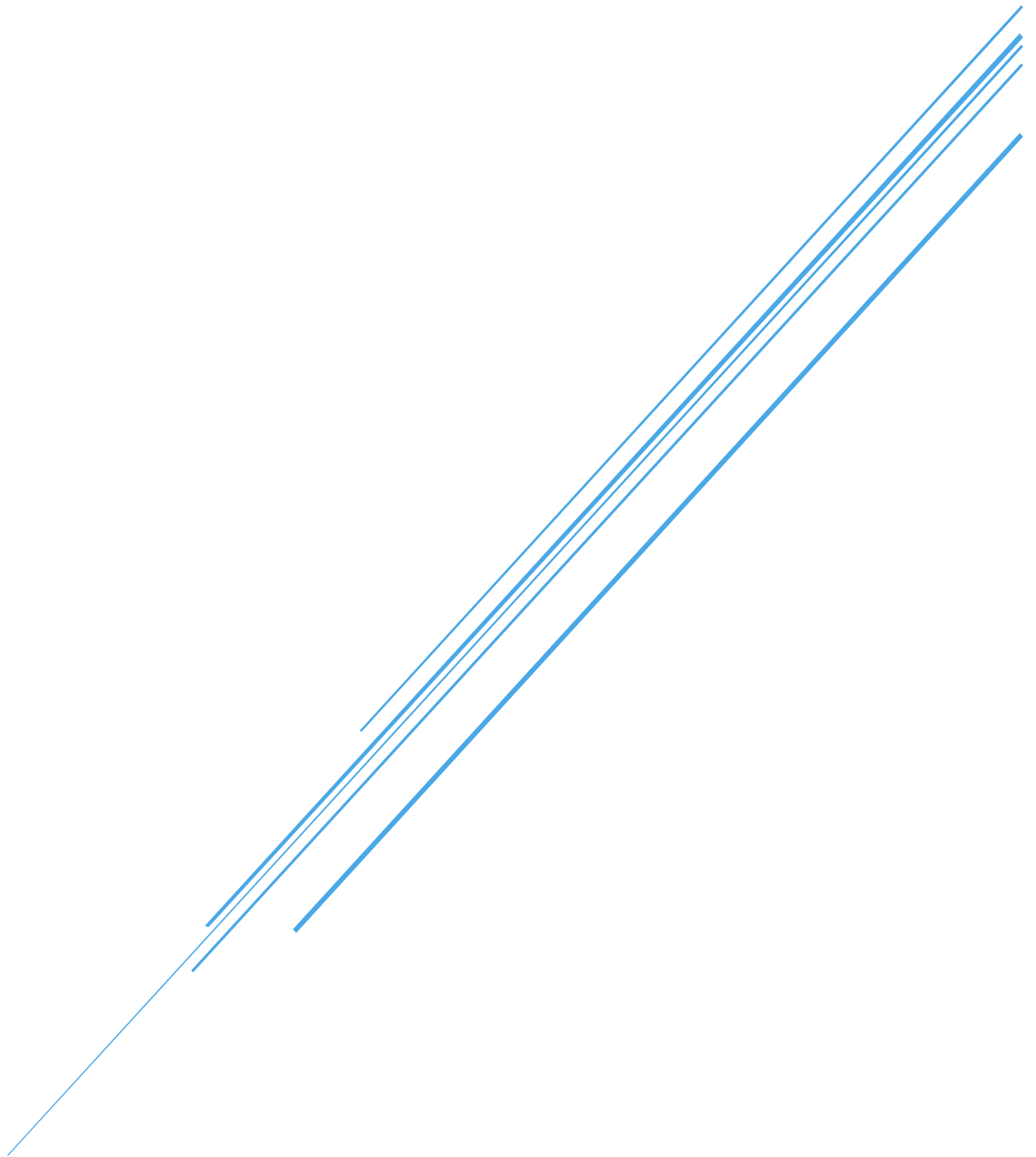


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CAPSTONE AND TEAM INTEGRATION STATEMENT

The overarching probabilistic system being built is a probabilistic graphical model for computing joint probabilities within energy sector and extending to attempting to estimate for certain latent states, such as energy stock price movements. Each team member has their own node which they own and are responsible for building an automated ETL style data pipeline to integrate into the larger model. All team members need to provide casual assumptions to turn these individual nodes into a connect larger graphical model (Directed Acyclic Graph). My respective node is the Geopolitical Risk (GPR) Index with the casual assumptions that GPR index influences the price of crude oil and the VIX Volatility Index [1]. My module integrates into the overall system by providing a directed edge from GPR Index to both crude oil and VIX nodes, allowing them to be used in computing joint probabilities. We have agreed on a data contract from the 1st of January 2018 to the 1st of January 2025, all nodes must adhere to this data contract, and each individual node owner is responsible for ensuring their subset of the dataset is clean and available.

PROJECT PROPOSAL

PROBLEM STATEMENT

Geopolitical events drive significant and often abrupt shifts in energy markets, affecting supply risk, prices, and operational planning. The issue this project addresses is that the underlying risk state relevant to energy markets is not directly observable; it can only be inferred indirectly through noisy indicators like GPR Index. This module therefore focuses on estimating geopolitical risk levels from the GPR Index and quantifying the uncertainty in those estimates, providing the probabilistic foundation required to later infer latent risk regimes and their transitions over time.

PROJECT OBJECTIVES

Below are the project objectives of the probabilistic machine learning engine in respect to my node in the PGM:

- Automated retrieval and cleaning of data.
- Bayesian vs Frequentist estimation comparison.
- Hypothetical DAG/PGM system integration
- Predicting for latent market regime states (Team component/Overarching DAG).

JUSTIFICATION FOR PROBABILISTIC APPROACH

The overarching engine having the goal of predicting for latent states relating to energy market stocks means that traditional machine learning (ML) approaches assume stability, once a function is fitted to data the assumption is that new data will also be similar. Markets can be notoriously volatile, and this assumption of stability can often break under extreme market conditions in which a traditional ML model has not been trained on [2]. This is why we need to quantify uncertainty; we need to know how ‘good’ of a guess our estimation is so it will allow us to make informed decisions and have a deeper understanding of the underlying risk of that decision.

MODEL SURVEY

The two main approaches for modelling uncertainty fall under two categories, Bayesian and Frequentist. Frequentist methods treat model parameters as fixed whereas Bayesian methods treat parameters as random variables with a prior, updated by the data into a posterior distribution.

A common Frequentist method is Maximum Likelihood Estimation (MLE), this method finds the fixed set of parameters that make the observed data most likely. This produces point estimates with confidence intervals.

A good choice for our Bayesian method is Markov Chain Monte Carlo (MCMC), this method uses random walks through the parameter space to sample the full posterior distribution over parameters. This gives uncertainty estimates instead of just single point values. Maximum A Posteriori is a method that gives the most probable posterior value, discarding the full distribution of uncertainty. MCMC is a more fitting choice due to its ‘full picture’ approach to the posterior distribution through random sampling.

HUMAN FACTORS AND ETHICS REVIEW

Traditional ML models give you the best guess they can without any uncertainty, in the world of financial markets a best guess without true confidence can result in massive losses, which in turn can have huge negative downstream repercussions. This can inherently lead to usage of models in this context displaying a level of false confidence that comes at a cost, not to also mention the un-interpretability of modern deep learning models and how hard it’s to figure out why a complex neural network gave a certain output.

Whilst a financial decision influenced by a model output may not put any human lives at risk it can certainly have an impact on someone’s livelihood. This is why being able to quantify uncertainty will allow people to make more informed financial decisions when interpreting model outputs.

STAKEHOLDER ANALYSIS AND CONFLICTING REQUIRMENTS

Two stakeholders of this system can be; a quantitative trader, and a regulatory body.

The quantitative trader requires a model they can trust the outputs from to make decisions on trades in short time windows with little and/or changing information. This means the system needs to be quick and responsive but also give outputs in a simple and interpretable manner.

The regulatory body wants model transparency for the sake of being able to explain and reason decisions made by model outputs to ensure compliance.

These requirements conflict directly. The trader prioritises low latency and a single actionable number, which favours the cheap MLE point estimate; the regulator prioritises auditable uncertainty and reasoning, which favours the fuller MCMC posterior at far higher computational cost. My design resolves this by running both pipelines and treating the MCMC posterior as the source of truth for transparency, while exploiting the fact that with 84 observations the posterior collapses onto the MLE (Table 1). This lets the trader safely use the fast estimate without sacrificing the regulator's auditability.

DATASET ANALYSIS

SOURCE

The data being used for this module is the Geopolitical Risk (GPR) Index (<https://www.matteoiacoviello.com/gpr.htm>). Which is a monthly risk index is computed by counting number of articles related to adverse geopolitical events in 10 newspapers monthly.

EXPLORATORY DATA ANALYSIS

The dataset is simple with their only being two columns (month, GPR Index score). The agreed upon data contract is from 1st January 2018 to 1st January 2025. This includes 84 rows of GPR Index scores with no missing values present.

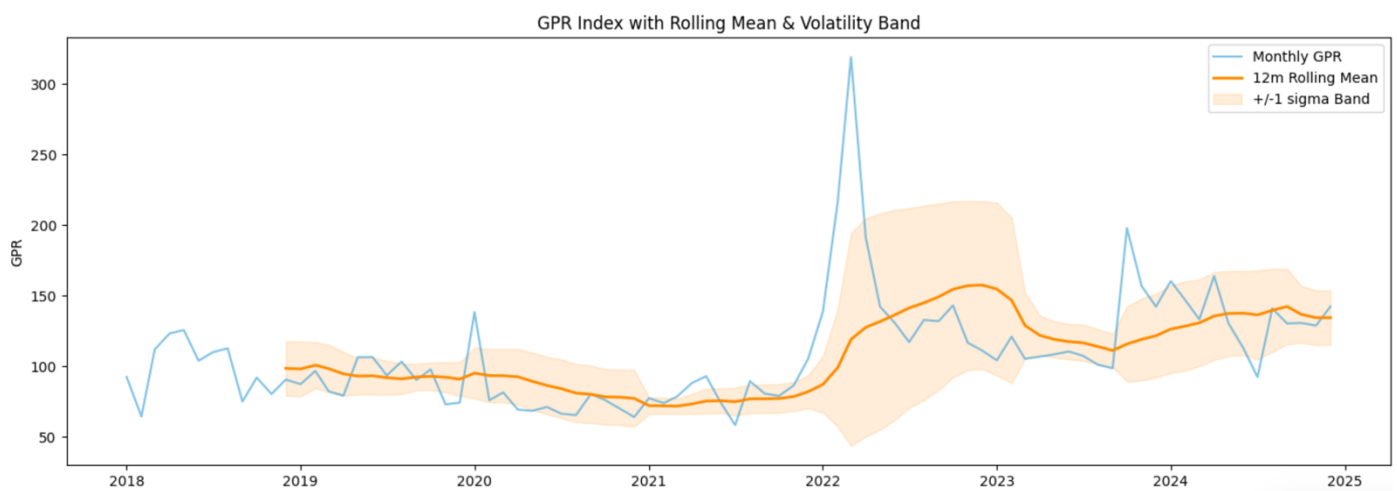


Figure 1 Rolling Mean with +/- 1 Sigma Band

Figure 1 shows the monthly GPR Index with a rolling mean and $\pm 1 \sigma$ volatility band. We can see that in early 2022 we have a huge spike in our GPR that falls way outside of that volatility band, likely due to some geopolitical events taking place at the time.

BAYESIAN VS FREQUENTIST COMPARISON

Parameter	Frequentist		Bayesian	
	MLE Estimate	MLE 95% Confidence Interval	Posterior	94% Highest Density Interval
μ	4.640	[4.574, 4.706]	4.640	[4.576, 4.702]
σ	0.308	[0.261, 0.355]	0.315	[0.269, 0.362]

Table 1 Frequentist vs Bayesian comparison for fitting lognormal distribution

Based on Table 1, we can see our Bayesian vs Frequentist approaches are nearly identical with the point estimates being the same up to 3 decimal places. Even the confidence interval and highest density intervals line up closely. This is likely because we have 84 months of data meaning the posterior gets heavily driven by likelihood essentially collapsing to MLE.

SYSTEM ARCHITECTURE AND IMPLEMENTATION

DESIGN AND PROCESS

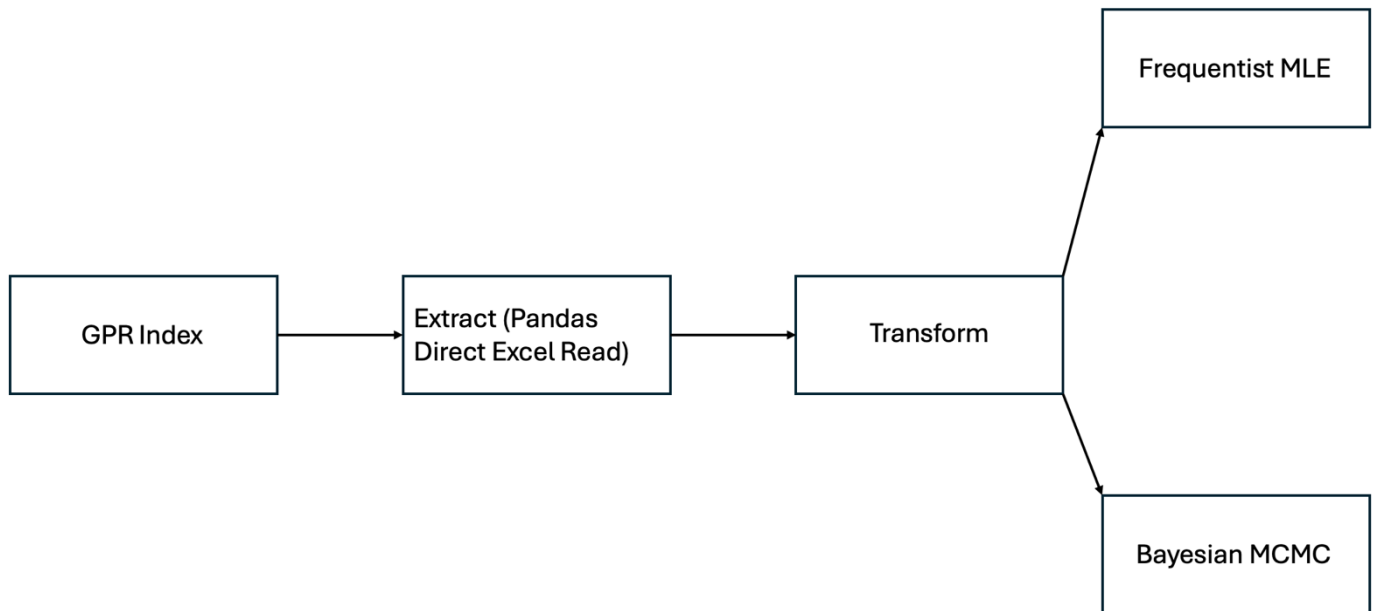


Figure 2 System Architecture Flow Chart

The following is derived from Figure 2:

- GPR Index: This is our data source.
- Extract: A using Pandas `read_excel()` function hits the excel endpoint from the GPR Index website. Directly reading in the excel file into a Pandas data frame object.
- Transform: We apply a lognormal transformation to the data (data is strictly positive and heavy right skewed).
- MLE: SciPy computes the MLE point estimates, Numpy computes the confidence intervals.
- MCMC: PyMC computes the posterior using No-U-Turn Sampler (NUTS) to take random walks through the parameter space. ArviZ uses the PyMC output to compute HDI 94% and R-hat.

TECHNICAL DECISIONS

The Python libraries used for this system are:

- Pandas: For handling the data and performing transformations.
- Numpy: For some linear algebra computations in MLE intervals. As well as being a dependency for almost every stat and ML Python library.
- SciPy: Frequentist MLE point estimate computations.
- PyMC: Bayesian MCMC posterior computations.
- ArviZ: Post-processing of MCMC; posterior mean computation, HDI 94%, and R-hat.
- Matplotlib/seaborn: Visualisation and rendering.

WORKS CITED

- [1] C. Behera , B. T. Priyadarsini and D. Patnaik, “Impact of Geopolitical Risk and Crude Oil Prices on Stock Return,” Issue Synergy and Innovation in Strengthening Resilience and Economic Revival, 2024.
- [2] S. A. Mamun, R. Hossain, J. Rahman, M. K. Devnath, F. Afroz and L. A. Amin, “Bayesian Modeling for Uncertainty Management in Financial Risk Forecasting and Compliance,” arXiv:2512.15739, 2025.
- [3] M. Iacoviello and D. Caldara, “Geopolitical Risk (GPR) Index,” 1 May 2026. [Online]. Available: <https://www.matteoiacoviello.com/gpr.htm>.

APPENDIX A: AI DECLARATION

Anthropic’s Claude Opus 4.7 (<https://claude.ai/new>) model was used for some explanation of PML concepts and brainstorming purposes. All code, report and video were not generated by any AI model.

Output is Summarised as it is quite lengthy, if full outputs are needed, they can be retrieved through chat logs with the model.

Prompt	Output Summary
Can you explain how MLE and MCMC work?	Maximum Likelihood Estimation (MLE) finds the single parameter setting θ that makes your observed data most probable by maximizing the (log-)likelihood $\Sigma \log p(x_i \theta)$, solved analytically or via gradient methods, yielding one best-fit point estimate without priors or a full account of uncertainty. Markov Chain Monte Carlo (MCMC) is a Bayesian method that instead approximates the entire posterior $p(\theta \text{data}) \propto p(\text{data} \theta)p(\theta)$, sidestepping the intractable normalizing constant by constructing a Markov chain whose stationary distribution is the posterior — e.g. Metropolis-Hastings proposes new parameters and accepts them with a probability where the constant cancels — and after discarding burn-in, the collected samples are summarized into point estimates, credible intervals, and distribution shapes. In short, MLE optimizes to one point and is cheap but ignores priors and uncertainty, while MCMC samples a full distribution, natively incorporating priors and quantifying uncertainty at greater computational cost.

What distributions fit heavy right skew with strictly positive values?	For strictly positive, right-skewed data, the standard candidates are: Log-normal — if $\log(x)$ is normally distributed. Very common default for multiplicative processes (income, biological measurements, particle sizes). Skew is governed by σ; can be extremely heavy-tailed. Gamma — flexible two-parameter (shape k, scale θ). Right-skewed when shape is small, approaches symmetric as shape grows. Good for waiting times, rainfall amounts, insurance claims. Lighter tail than log-normal. Weibull — two-parameter, popular for survival/reliability and lifetimes. Shape < 1 gives strong right skew and a decreasing hazard; shape > 1 can reduce skew. More tail flexibility than gamma. Exponential — single-parameter special case of both gamma ($k=1$) and Weibull (shape=1). Memoryless; fits inter-arrival times. Fixed skewness of 2, so only fits if the data's skew matches. Pareto / power-law — for very heavy tails where a few extreme values dominate (wealth, city sizes, file sizes). Polynomial rather than exponential tail decay; may have infinite variance or even infinite mean. Inverse Gaussian — right-skewed, used in reliability and as the basis for some GLMs. Heavier-tailed than gamma for comparable parameters. F and chi-squared — positive and right-skewed, but usually arise as test statistics rather than as models you'd fit to raw data.
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Table 2 AI Prompts